

## THE FURST OPTICAL PERFORMANCE AND RESPONSE SECTIONS

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*Draft version October 9, 2026*

### ABSTRACT

This document is not the FURST instrument paper. It is a preview of the sections of it which are generated from the instrument model, built so that they can be read and checked on their own before they are delivered to the manuscript. The numbered subsections above them are empty placeholders, standing in for the sections of the manuscript which precede them.

#### 1. INSTRUMENT OVERVIEW

##### 1.1. *Design Rationale*

##### 1.2. *Optical Layout*

##### 1.3. *Optical Performance*

The performance of the instrument as it was built and flown was evaluated by tracing rays through a model of it, `furst-optics` (Smart 2026b), built with the open-source `optika` raytracing package (Smart and Kankelborg 2026). The model is shown in Figure 1 and its parameters are listed in Table 1. It places the seven feed optics, the grating, and the detector on a Rowland circle of radius 677 mm, as the instrument was laid out for a grating with a radius of curvature of 1354 mm. The flight grating came out with a radius of 1359 mm, but the grating and the detector were left where the layout put them, so the model keeps them there and changes only the curvature of the grating. Its ruled area is 181.9 mm wide and 36 mm tall, with a ruling density of 2200 mm<sup>-1</sup>, as Zeiss measured them for its final report on the grating (Stock 2023). The Sun is the field stop of the system and the grating is its pupil stop: the field of view is the solar disk at its largest apparent size, and every ray leaving a point on the disk is directed at a point on the ruled area of the grating. Each feed optic is a convex cylinder of radius 3 mm whose clear aperture subtends 45° of the cylinder and is 20.8 mm tall.

The feed optics illuminate the grating at angles of incidence from 10.42° to 17.82°, and the detector lies 4.99° from the grating normal on the same side, so that the shortest wavelengths are observed close to the Littrow configuration. Each channel therefore places a band about 10.3 nm wide on the detector, and the seven bands together cover 116.6 nm to 183.8 nm, with adjacent channels overlapping by at least 0.7 nm so that the spectrum can be stitched together. The dispersion at the detector is 5.02 pm per pixel.

A visible-blind filter on a 2.055 mm window of magnesium fluoride sits in front of the detector and moves the focus behind it, and the longer radius of the flight grating moves it farther still. As on the bench, the model brings the instrument back into focus by moving the feed optic array alone: it slides the array 11 mm along the axis of the instrument, which focuses the first channel, and piv-

ots it 0.19° about the first feed optic, which focuses the last.

Since the virtual image of the entire Sun formed by a feed optic is narrower than a pixel, the quantity that governs the spectral resolution is the disk-integrated line spread function (LSF): the distribution along the dispersion direction of the light from the whole solar disk and the whole pupil at a single wavelength. We compute it by tracing 5 wavelengths spanning the detector in each channel, with the solar disk sampled by a stratified random grid of 15 × 15 points and the pupil by one of 15 × 15 points. The result is shown in Figure 2: the image of the Sun at each wavelength is a line about a pixel wide, taller than the detector, and its profile along the dispersion direction is roughly Gaussian. We take the width of the LSF to be the standard deviation along the dispersion direction of the positions of the rays reaching the detector, added in quadrature with the standard deviation of a uniform distribution one pixel wide, to account for the finite size of the pixels. This width is shown in the top panel of Figure 3 and ranges from 0.40 to 0.67 pixels. It is largest in the middle channels: since the radius of the flight grating differs from that of the layout, only the two end channels can be brought to their best focus, one by each motion of the feed optic array, and the channels between them sit a little off it.

The resolving power, taken as the wavelength divided by twice the width of the LSF, is shown in the bottom panel of Figure 3. It ranges from 21,600 to 36,300 over the traced wavelengths, with a mean of 29,700, and so exceeds the requirement of  $2 \times 10^4$  throughout the bandpass.

Every figure in this section, and every number it takes from the model, is computed from the instrument model by `furst-instrument-paper` (Smart 2026a), so that they cannot drift from it.

##### 1.4. *Response*

The response of the instrument is the rate at which it records electrons for a given irradiance at its entrance. We compute it from the same model of the instrument and show it, with each of its factors, in Figure 4: the efficiency of each optic the light meets, and the quantum efficiency of the detector. Each is computed one channel at a time, since each channel meets the grating at its own

TABLE 1

OPTICAL DESIGN PARAMETERS OF THE INSTRUMENT MODEL USED IN THIS SECTION, BESIDE THE VALUES IN TABLE 3 OF THE DRAFT OF SEPTEMBER 18, 2026. THE GRATING OF THE MODEL IS THE FLIGHT GRATING, WHOSE RULED AREA AND RULING DENSITY ARE AS ZEISS MEASURED THEM (STOCK 2023). ENTRIES IN BOLD DIFFER BETWEEN THE TWO AND NEED TO BE RECONCILED BEFORE SUBMISSION.

| Parameter                                  | Draft                 | Model                                  |
|--------------------------------------------|-----------------------|----------------------------------------|
| Wavelength range                           | 116 to 185 nm         | 116.6 nm to 183.8 nm                   |
| Resolving power                            | > 20,000              | 21,600 to 36,300                       |
| Number of feed optics                      | 7                     | 7                                      |
| Radius of feed optic                       | 3 mm                  | 3 mm                                   |
| Height of feed optic                       | <b>38 mm</b>          | 20.8 mm ( <b>clear aperture</b> )      |
| Feed optic positions (from grating normal) | 10.5° to 17.9°        | 10.42° to 17.82°                       |
| Radius of grating                          | 1359 mm               | 1359 mm                                |
| UV section of grating                      | <b>200 mm × 30 mm</b> | 181.9 mm × 36 mm ( <b>ruled area</b> ) |
| Groove density                             | <b>1.97 gr/μm</b>     | 2200 mm <sup>-1</sup>                  |
| Detector type                              | CCD230-42             | CCD230-42                              |
| Detector format                            | <b>2048 × 1024</b>    | <b>2048 × 1032</b>                     |
| Pixel size                                 | 15 μm                 | 15 μm                                  |
| CCD position (from grating normal)         | 4.98°                 | 4.99°                                  |

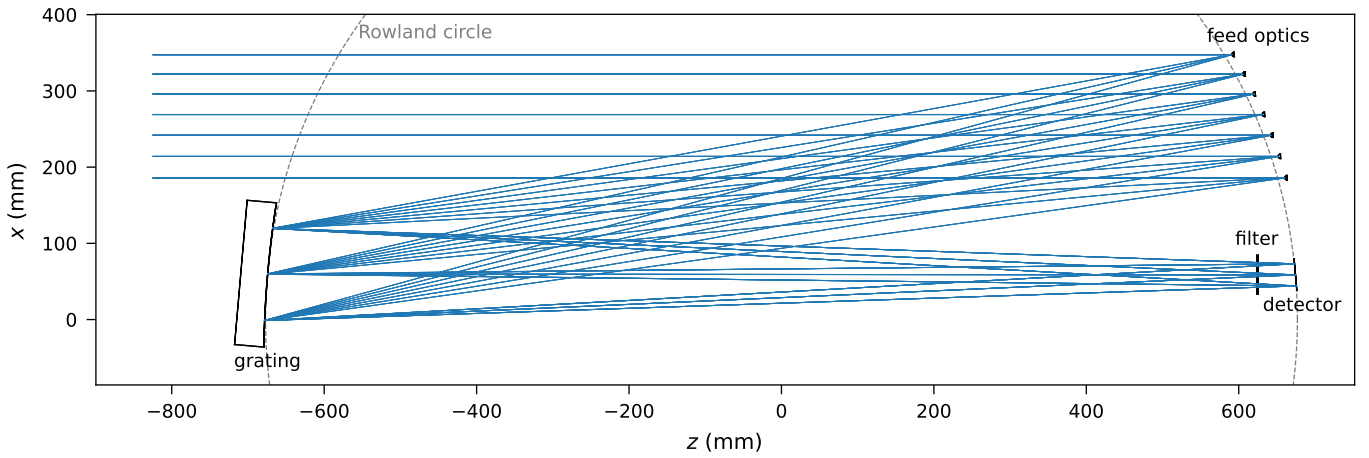


FIG. 1.— The optical layout of FURST, seen from above the optical table. Sunlight enters from the left and is reflected by one of seven convex cylindrical feed optics, each of which forms a virtual image of the Sun on the Rowland circle (dashed), narrower than a pixel of the detector. The concave diffraction grating, at the far side of the circle, disperses the light and focuses it back onto the circle at the detector, through a visible-blind filter just in front of it. Each feed optic illuminates the grating at a different angle of incidence, so each channel places a different band of the spectrum on the detector. Three wavelengths are traced through each channel.

angle, and the neighboring channels agree where their bands overlap.

The feed optics reflect 79% of the light on average, as measured on witness samples coated alongside them. The grating sends 29% of it into the first order: the reflectance Zeiss measured on a test piece coated with it, times the efficiency Zeiss computed for grooves of the profile it measured on the flight grating (Stock 2023), at the angle of incidence of each channel. The visible-blind filter, as measured on its witness sample, transmits 20% on average, and less toward Lyman  $\alpha$ .

The quantum efficiency of the detector, in electrons per incident photon, is the product of three factors: the fraction of the photons absorbed in its silicon, 36% on average; the fraction of the charge they liberate which reaches a pixel, its charge collection efficiency, 0.36; and the number of electron-hole pairs each absorbed photon liberates, its quantum yield, which falls from 3.0 to 1.7 across the bandpass. We do not correct it for the quantum yield, so it counts electrons rather than photons, and it ranges from  $0.22 \text{ e}^- \text{ photon}^{-1}$  to  $0.38 \text{ e}^- \text{ photon}^{-1}$ . It is the least certain of the factors, since the detector is modeled rather than measured: the charge collection efficiency is that of a model of an e2v CCD97 fit in part

to the measurements of Heymes et al. (2020), not of the CCD230-42 which flew, and the quantum yield is the model of Ramanathan and Kurinsky (2020), which is fit to measurements only at photon energies below 8 eV, longward of 155 nm.

The response, in the bottom panel of Figure 4, is the product of these factors and the area the instrument collects light from, divided by the energy of a photon. That area,  $4.1 \text{ mm}^2$ , is set by the layout alone, and is small because each convex feed optic spreads the light it reflects over a wide fan, of which the grating intercepts only a small part. Multiplied by the irradiance of a spectral line in  $\text{erg cm}^{-2} \text{ s}^{-1}$ , the response gives the electrons per second that the line deposits on the detector. It rises steeply across the first channel, from  $1.3 \times 10^7 \text{ e}^- \text{ cm}^2 \text{ erg}^{-1}$ , as the transmission of the filter and the absorbance of the detector both fall toward Lyman  $\alpha$ , and reaches  $5.2 \times 10^7 \text{ e}^- \text{ cm}^2 \text{ erg}^{-1}$  in the middle of the bandpass. Since the quantum yield falls almost in proportion to the energy of a photon, the detector records nearly the same charge for each unit of energy it absorbs, and beyond the first channel the response follows the product of the efficiencies of the three optics. Table 2 sets each of these factors beside the estimates in

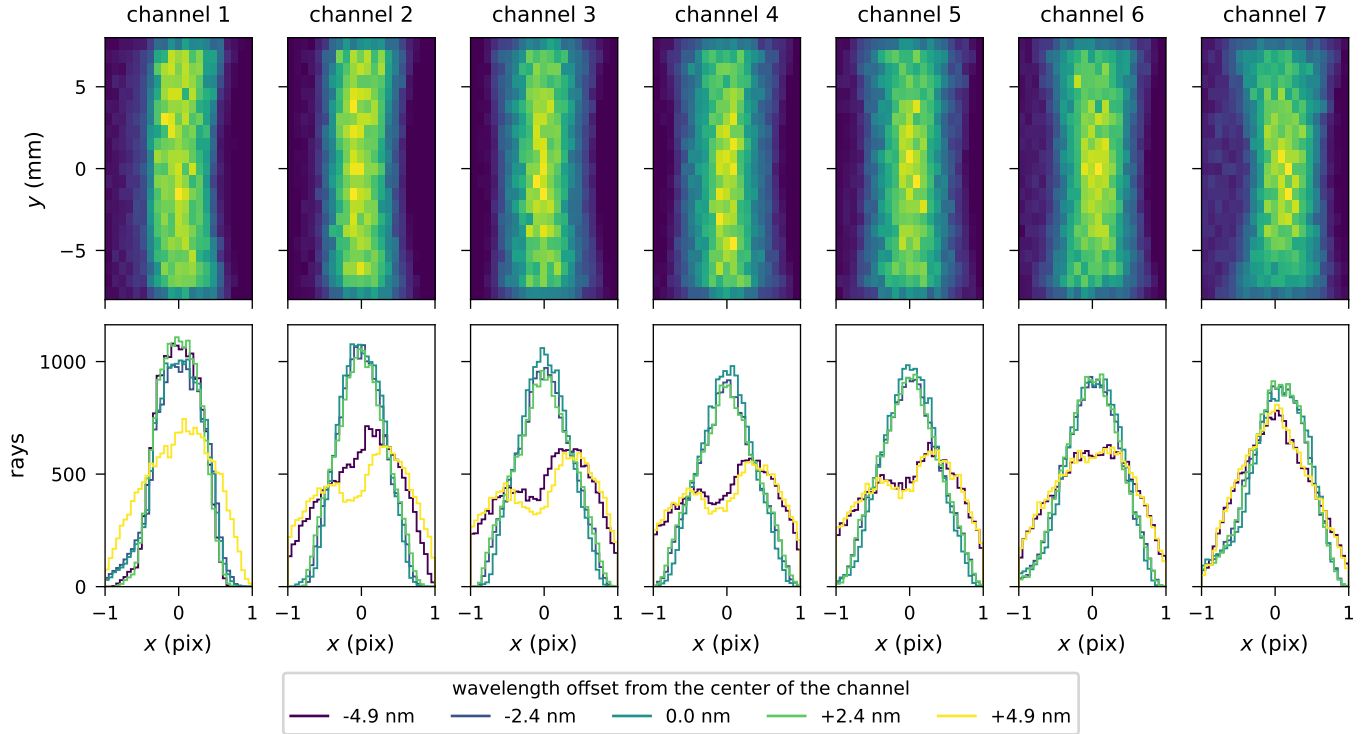


FIG. 2.— The disk-integrated line spread function of each channel of FURST, from a raytrace of the instrument model. Top: the distribution of the rays on the detector at the central traced wavelength of each channel, with the dispersion direction,  $x$ , measured in pixels from the mean position of that wavelength and the height on the detector,  $y$ , measured from its center. Bottom: the same distribution summed along the height of the detector, at every traced wavelength of each channel, from the shortest (dark) to the longest (light).

Table 4 of the draft.

TABLE 2

TERMS OF THE EFFECTIVE AREA IN TABLE 4 OF THE DRAFT OF SEPTEMBER 18, 2026, BESIDE THOSE OF THE MODEL, AVERAGED OVER THE SAMPLED WAVELENGTHS OF EVERY CHANNEL, EXCEPT FOR THE EFFECTIVE AREA AND THE QUANTUM YIELD, WHICH ARE GIVEN AS THEIR RANGE. AS IN THE DRAFT, THE EFFECTIVE AREA IS THE PRODUCT OF THE FIVE TERMS ABOVE IT. THE QUANTUM EFFICIENCY OF THE MODEL IS COUNTED IN ELECTRONS PER PHOTON, SO IT INCLUDES THE QUANTUM YIELD, WHICH THE DRAFT TAKES TO BE ONE IN ITS TABLE 5.

| Term                | Draft                  | Model                                          |
|---------------------|------------------------|------------------------------------------------|
| Geometric area      | 3.072 mm <sup>2</sup>  | 4.1 mm <sup>2</sup>                            |
| Mirror reflectivity | 85%                    | 79%                                            |
| Grating efficiency  | 30%                    | 29%                                            |
| Filter transmission | 13%                    | 20%                                            |
| Quantum efficiency  | 20%                    | 0.30 e <sup>-</sup> photon <sup>-1</sup>       |
| Effective area      | 0.0204 mm <sup>2</sup> | 0.022 mm <sup>2</sup> to 0.075 mm <sup>2</sup> |
| Quantum yield       | 1                      | 1.7 to 3.0                                     |

## REFERENCES

- Julian Heymes, Matthew Soman, George Randall, Alexander Gottwald, Andrew Harris, Andrew Kelt, Ian Moody, Xiao Meng, and Andrew D. Holland. Comparison of Back-Thinned Detector Ultraviolet Quantum Efficiency for Two Commercially Available Passivation Treatments. *IEEE Transactions on Nuclear Science*, 67(8):1962–1967, 2020. doi:10.1109/TNS.2020.3001622.
- K. Ramanathan and N. Kurinsky. Ionization yield in silicon for eV-scale electron-recoil processes. *Phys. Rev. D*, 102(6):063026, 2020. doi:10.1103/PhysRevD.102.063026.
- Roy T. Smart. first-instrument-paper, 2026a. URL <https://github.com/roytsmart/furst-instrument-paper>.
- Roy T. Smart. first-optics, 2026b. URL <https://furst-optics.readthedocs.io>.
- Roy T. Smart and Charles C. Kankelborg. Optika, 2026. URL <https://optika.readthedocs.io>.
- Carsten Stock. FURST FUV Grating: Final Report. Technical report, Carl Zeiss Jena GmbH, July 2023.

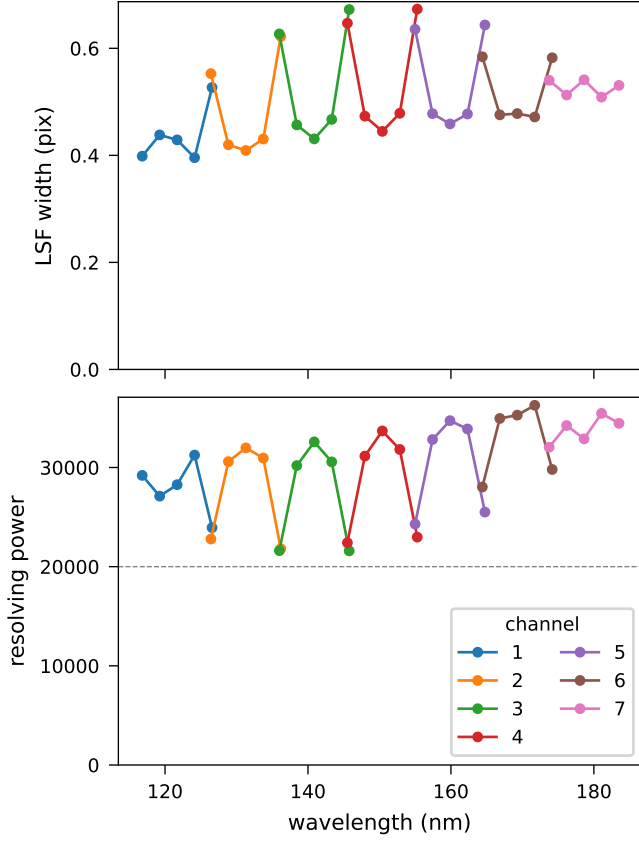


FIG. 3.— The spectral resolution of each channel of FURST at the traced wavelengths, computed from a raytrace of the instrument model. Top: the width of the disk-integrated line spread function, the standard deviation of the positions of the rays along the dispersion direction, added in quadrature with the standard deviation of a uniform distribution one pixel wide. Bottom: the resolving power, taking the resolvable wavelength difference to be twice that width. The dashed line marks the requirement of  $2 \times 10^4$ .

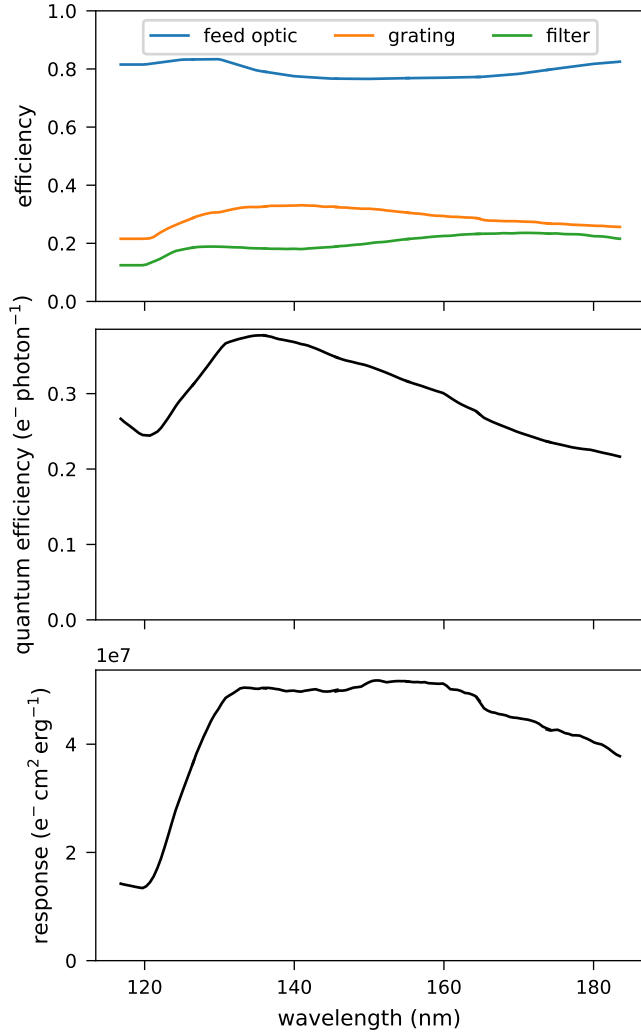


FIG. 4.— The response of FURST and its factors, from the instrument model, drawn one channel at a time. Top: the reflectance of the feed optics, the first-order efficiency of the grating at the angle of incidence of each channel, and the transmission of the visible-blind filter. Middle: the quantum efficiency of the detector, in electrons per incident photon, the product of its absorbance, its charge collection efficiency, and its quantum yield. Bottom: the response, the electrons recorded for each erg per square centimeter reaching the instrument, averaged over the solar disk. Multiplied by the irradiance of a spectral line in  $\text{erg cm}^{-2} \text{ s}^{-1}$ , it gives the electrons per second that the line deposits on the detector.